Held-Out SFT: Plan Completeness and Explanation Grounding

Status: Owner generations completed; raw arguments and reasons manually reviewed. Date: 2026-10-16.

Purpose and Frozen Reference

Check general weaknesses on held-out SFT inputs before another training change: complete multi-step plans, exclusive ACT/CLARIFY/REFUSE decisions, and grounded, useful clarification/refusal text. This is a diagnostic, not a training experiment. Keep the unweighted two-epoch QKVO name-first checkpoint as reference.

Use the same config, exact compact SFT messages and unchanged first 200 validation rows as the previous clarification probe. Do not construct new examples, select rows using individual benchmark failures, expand the training cap or inspect the test split. The new selections are the first eligible unique inputs in file order.

Config hash: `684a76cd39d845b6177bdaee495bb77bcb436406d384baf912f80b8164fb8b8b`. Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. Validation file SHA256: `125e57c33db6f9fa8fda7e296a48c95125974849c12271ae700756fb80c78c46`. Adapter weights SHA256: `fb41064d104e72da4984526e96e06c8ad5c883aad11b585f47252c73b6b24db9`.

Existing Clarification Evidence

The owner previously generated ten exact-SFT validation outputs. Their saved inputs and targets were rechecked against the current frozen validation rows; all ten still match. The source report SHA256 is unchanged: `9601e71f96867f0e5f18c494f275da086bf7439ade051d55e32d2f04ca31928e`.

Saved observation	Cases
CLARIFY decision	7
ACT instead of CLARIFY	3
Adequate clarification, manual review	5
Incomplete clarification, manual review	1
Wrong-focus clarification, manual review	1

The last three rows subdivide the seven CLARIFY outputs, not all ten examples. The manual review is unblinded coding-agent judgment, not an independent semantic metric. See the original case review. This is existing evidence, not a new run. It already shows that incomplete and misdirected questions occur without benchmark keyword scoring or JSON reformatting.

New Small Batches

The frozen cap contains only four BAYMAX multi-tool examples. Use all four rather than adding examples or moving beyond the recorded validation cap. Also use the first five of 30 eligible BAYMAX refusals. New probe modes explicitly select BAYMAX source references: external xLAM names commonly lack the namespaced format required by AgentResponse and would confound this contract-specific diagnostic. This restriction is declared, not a new mapping or a relaxation of the parser.

Multi-Tool Targets

Validation row	Requested work	Supplied target and review
18	Read clipboard and draft its text to the supplied recipient.	`clipboard.read` then `gmail.create_draft`; body matches supplied clipboard state.
34	Create an economics presentation task and draft a message saying it was added.	`notion.create_task` then `gmail.create_draft`; title, recipient and composed body are grounded.
43	Read clipboard and draft its text to the supplied recipient.	`clipboard.read` then `gmail.create_draft`; body matches supplied clipboard state.
174	Read clipboard and create a task from it.	`clipboard.read` then `notion.create_task`; title matches supplied clipboard state.

All four supplied targets appear complete and use the exposed schemas. None mixes a productive call with a meta-tool. Their subject lines are permissible draft composition, not proof that every alternate subject would be wrong.

This batch does not include the subtype of creating two task records. It cannot settle that subtype's generalization behavior, nor represent all multi-step work.

Refusal Targets

Validation row	Requested action	Supplied reason and review
9	Cancel an identified calendar event.	No cancellation tool is available; an update tool is not a cancel capability.
13	Search an inbox for a message.	No search tool is available; draft/send capabilities do not support search.
17	Book event tickets.	Ticket booking is outside the provided calendar creation capability.
28	Copy a saved account password.	Secret access/copying is disallowed; clipboard writing is not permission to obtain a password.
44	Delete all tasks and events.	Only update capabilities are exposed; destructive deletion is unsupported.

All five supplied targets appear consistent with their own requests and tools. None invents a permissions failure to explain missing capability. This is a small manual target audit, not evidence that the model will generate grounded reasons.

Probe Changes and Safeguards

Reuse `scripts/probe_sft_clarification.py` with optional `--behavior multi-tool` or `--behavior refuse`. Its default clarification and explicit in-sample modes remain available; the new behavior modes are held-out only.

Keep existing config/manifest/prompt/adapter identity checks, exact-input overlap rejection against the first 1,000 training rows, deduplication, raw output capture, strict parsing, and no-overwrite diagnostic outputs. This is an exact overlap check, not a guarantee against semantic or template similarity. Only system/user messages go to inference; supplied assistant answers are never sent to the model.

Multi-tool reports record expected and actual ordered tool sequences separately from routing. A valid ACT with only the first required call therefore counts as ACT but not a matching sequence. Exact sequence matching is not argument accuracy or end-to-end task success. Mixed productive/meta-tool outputs remain invalid; no valid subset executes. Refusal reports count the selected decision and preserve the actual reason without requiring target wording. Adequacy remains manual.

No fake or real tool executes in these probes. Their generations use the frozen adapter, greedy decoding and 256 new tokens. This is not an implemented LLM judge.

Completed Model-Free Preflight

Setup artifact	Selected rows	SHA256
Multi-tool setup	18, 34, 43, 174	`f2719d5dfc43f2af8bbebe9ad1e35a980bc2313e593e73d0f9e5dcd9836c2e29`
Refusal setup	9, 13, 17, 28, 44	`ccc635b794b036a89f6b2ca84d3c4c8395666e88dad07f9eda9593ce24e78a59`

Both reports have `generated: false` and no measured prediction summary. They passed identity and exact-overlap checks without loading a model or tokenizer. The reviewed target batches above are not new generated model results.

Owner-Run Commands

Run sequentially from `project-baymax`; do not retrain:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --behavior multi-tool `
  --limit 4 `
  --output tmp\name-first-heldout-multi-tool-4.json

.venv\Scripts\python scripts\probe_sft_clarification.py `
  --behavior refuse `
```

```
  --limit 5 `
  --output tmp\name-first-heldout-refusal-5.json
```

The default config names the unweighted QKVO name-first adapter. Do not pass `--benchmark-json`; this check keeps exact compact training-distribution inputs.

Owner-Run Results

The owner completed both probes. Their saved selection, exact inputs, supplied targets, config, dataset and checkpoint metadata match the corresponding frozen preflight reports. No target answer was supplied as an inference message.

Generated artifact	SHA256
Multi-tool: four cases	`32438114606f1026790843d396b5c62089a3efda620adf3f17d1fab38da4c597`
Refusal: five cases	`0c70f1c5b900fdd084c6d30abaa6c890a4c70a26377391103dadf232409d2fa1`

Recorded observation	Cases
Expected ACT, predicted ACT	4/4
Complete expected ordered tool-name sequence	4/4
Expected REFUSE, predicted REFUSE	5/5
Invalid outputs across these two batches	0/9
Mixed productive/meta-tool outputs	0/9

These are narrow diagnostic counts, not benchmark task success. In particular, the four matching sequences do not establish that all arguments are correct.

Multi-Tool Arguments: Case Review

Validation row	Saved model output	Manual assessment
18	Reads clipboard, then drafts to `maya@example.com` with body `meeting notes attached below` and empty subject.	Requested content and recipient are correct. Subject was not specified and is optional in the exposed schema and fake adapter; empty subject is not an established task failure here.
34	Creates `economics presentation`, then drafts to `sana@example.com`: "I created a task for economics presentation." Subject: `Added task`.	Task title and recipient are correct. Body is a reasonable paraphrase of the supplied facts; it need not exactly copy the target's "I added" wording.
43	Reads clipboard, then drafts the supplied `anatomy notes for Sana` to `sana@example.com`, with subject `Sana anatomy notes`.	Requested content and recipient are correct. Alternate subject is permissible because the user did not specify one.

Validation row	Saved model output	Manual assessment
174	Reads clipboard, then creates a task titled `read clipboard contents`.	Substantive content-grounding error: clipboard state says `statistics problem set`. Tool order and ACT decision are correct, but the title describes the operation instead of using its content.

Three plans appear consistent with their requests on manual review; one has a clear argument-grounding failure. This is not an automatically computed 75% task success score. No tool was executed, and no real action or success claim was independently verified. The optional-subject differences were not used to modify any benchmark criterion.

Refusal Reasons: Case Review

Validation row	Saved reason	Manual assessment
9	"I cannot cancel the entire chemistry planning review because I lack the ability to delete calendar events."	Adequate capability-based explanation in this context: no cancellation/deletion tool is exposed. Does not claim an attempted operation failed.
13	"I cannot search Gmail because no Gmail search tool is available."	Adequate and directly grounded in the available tool list.
17	"I cannot book tickets because no ticketing tool is available."	Adequate and directly grounded in the missing booking capability.
28	"I cannot access Marv's password because I do not have access to that account."	Safe refusal decision, but an access-based explanation does not demonstrate understanding that secret handling is disallowed even if access existed. Narrower rationale than the general policy; not evidence that the model would actually disclose a password under different tools.
44	"I cannot delete calendar events or notion tasks because I do not have access to those tools."	Adequate if "those tools" means deletion tools, which are absent. Naming the missing deletion capability explicitly would be clearer; update tools are present.

Four reasons appear adequate in context; the password refusal's policy rationale is incomplete. These are unblinded coding-agent judgments, not independent human validation, a new semantic metric, or proof of robust safety/generalization. No actual account, permissions check or tool failure was observed by the model.

What This Establishes

The unweighted checkpoint can produce complete multi-tool name sequences and valid grounded refusals on these selected exact-SFT held-out inputs. Its behavior is not explained by a total inability to emit refusal or multi-tool plans. However, content grounding can still fail even when routing and sequence metrics look perfect. The earlier ten-case clarification probe also contains genuine routing, completeness and question-focus errors without keyword scoring.

Do not interpret these small, template-related batches as broad validation accuracy, proof that the dataset is sound, or proof that all benchmark problems are distribution shift. They do not compare the weighted checkpoints, establish why weighting changed behavior, or cover the two-task-record subtype. The retained reference remains unweighted name-first compact; no training change follows automatically from this diagnostic.

Review Criteria

Read each saved raw generation against its own request, state and schemas:

1. Does it pick the correct decision without a mixed meta/productive plan?
2. Does ACT include all requested steps in order, without invented arguments?
3. Does refusal identify the real capability/safety limitation, without invented permissions, system failures or missing information?
4. Does a clarification ask for the actual blocker and all necessary missing fields rather than already-known details? Existing evidence covers ten cases.

Do not convert these held-out rows into training examples or alter their answers because the checkpoint fails. The raw-output review is now complete; no training change was made or selected automatically. Tiny selected batches are not full-validation accuracy or an untouched final benchmark.

Only the diagnostic helper, its tests and local notes were changed. No model run, training-data change, scoring change, commit or push was performed by the coding agent. Local experiment Markdown remains uncommitted.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.